

G. Lejeune Dirichlet's Works. II.

I.

On the Stability of Equilibrium.

By G. Lejeune Dirichlet.

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On the Stability of Equilibrium.

[Extract from a paper read in the Royal Academy of Sciences on 22 January 1846.]¹

When attractive or repulsive forces, depending only on distance, act on a system of material points, either directed toward fixed centers, or such that, when they occur mutually between two masses, action and reaction are equal; and when, moreover, the constraint equations connecting the coordinates of the points with one another do not contain the time, then the so-called integral of vis viva always holds. In its full generality it was first set out by D. Bernoulli, and is contained in the equation

$$\Sigma mv^2 = f(x, y, z, x', \dots) + C.$$

The summation sign extends over all masses of the system; each is denoted by m , its velocity by v , and C denotes the arbitrary constant. The function of the coordinates depends only on the nature of the forces and can always be expressed by a definite number of independent variables $\lambda, \mu, \nu, \dots$, whereby the equation becomes

$$\Sigma mv^2 = \varphi(\lambda, \mu, \nu, \dots) + C.$$

The function φ is intimately related to the equilibrium positions of the system, since the condition that equilibrium should take place for a position corresponding to specified values of $\lambda, \mu, \nu, \dots$ coincides with the vanishing of the total differential of φ for these values; thus, in general, for every equilibrium position the function will have a maximum or a minimum. If a maximum actually occurs, then the equilibrium has the character of stability; that is, if the points are displaced only slightly from such a position and receive small initial velocities, the system will never, in the course of time, move away from that position beyond certain narrow bounds.

The theorem just mentioned is unquestionably among the most important in all mechanics; in particular, the theory of small oscillations, which contains so many interesting physical applications, rests upon it. One must indeed wonder that its truth has not until now been demonstrated with the necessary rigor. Suppose that the equilibrium position, or the maximum of the function φ , corresponds to the values $\lambda = 0, \mu = 0, \dots$; this can always be done without loss of generality. The proof given by Lagrange² consists in expanding the function in powers of $\lambda, \mu, \nu, \dots$; since this expansion begins with terms of the second order, he restricts it to those terms, and then from the known criterion for a maximum, according to which the second-order terms can be represented as a sum of negative squares, derives bounds for $\lambda, \mu, \nu, \dots$ which they cannot exceed. This line of reasoning, which is also not seldom applied to other questions of stability, especially in physical astronomy, is plainly without demonstrative force. It can rightly be doubted whether quantities for which, under the supposition that they always remain small — for only in this supposition lies the permission to neglect the higher terms —, one finds small

¹In the 1846 volume of the Academy reports the extract is introduced on p. 34 with the words: Mr. Lejeune Dirichlet read *on the conditions for the stability of equilibrium*. For the printing in the Journal, Dirichlet changed the title, as well as the text in a few places, and newly added the entire final paragraph. K.

²Mécanique analytique, première partie, section III.

bounds, will after an arbitrary time actually be enclosed within these, or even within narrow bounds at all.

So far as I know, the proof just mentioned has hitherto been reproduced without essential alteration by all writers who have dealt with this subject; and what Poisson³ in particular added to it, in order to take account of the terms of higher order, rests on the quite untenable assumption that each term of the second order exceeds the aggregate of all higher terms. But even if Lagrange's considerations were completed for the case to which they apply, and in which the maximum is revealed in the terms of second order, the theorem in question would still not thereby be proved in its full extent. As is well known, the existence of a maximum is compatible with the vanishing of the second-order terms; only the lowest order for which not all terms vanish must always be even, and the aggregate of all terms of that order must always remain negative. The complete criteria for this latter condition have not yet been established, even when the fourth order is concerned (*Théorie des fonctions, seconde partie, no. 57*);⁴ they would therefore have to be sought first, which would necessarily introduce great complication into the proof of the mechanical theorem.

Fortunately, the principle of the stability of equilibrium can be established independently of these criteria, and by very simple considerations directly connected with the concept of a maximum.

Keeping the above supposition that the equilibrium position corresponds to vanishing values of $\lambda, \mu, \nu, \dots$, we make the second supposition that the value of $\varphi(0, 0, 0, \dots)$ is likewise zero; this is allowed because of the arbitrary constant. If one now determines the constant from the given initial state, for which $v, \lambda, \mu, \nu, \dots$ are to be denoted by $v_0, \lambda_0, \mu_0, \nu_0, \dots$, one obtains

$$\Sigma mv^2 = \varphi(\lambda, \mu, \nu, \dots) - \varphi(\lambda_0, \mu_0, \nu_0, \dots) + \Sigma mv_0^2.$$

By the assumption that $\varphi(\lambda, \mu, \nu, \dots)$ has, for $\lambda = 0, \mu = 0, \nu = 0, \dots$, a maximum whose value itself vanishes, one can always specify such small positive quantities $l, m, n, \dots$ that for every system $\lambda, \mu, \nu, \dots$, if the numerical values of these variables are respectively no greater than $l, m, n, \dots$, $\varphi(\lambda, \mu, \nu, \dots)$ is always negative, except in the single case where $\lambda, \mu, \nu, \dots$ vanish simultaneously. This case is excluded if we consider only such systems in which at least one of the variables $\lambda, \mu, \nu, \dots$, apart from sign, is equal to its bound $l, m, n, \dots$. If, among all these negative values of the function, $-p$ is the smallest apart from sign, it is easy to show that, if $\lambda_0, \mu_0, \nu_0, \dots$ are taken numerically smaller than $l, m, n, \dots$ and at the same time the condition

$$-\varphi(\lambda_0, \mu_0, \nu_0, \dots) + \Sigma mv_0^2 < p$$

is satisfied, then each of the variables $\lambda, \mu, \nu, \dots$ will remain, in the course of the motion, numerically below its bound $l, m, n, \dots$. Were the contrary to occur, then, since the initial values $\lambda_0, \mu_0, \nu_0, \dots$ satisfy the condition just stated by hypothesis, and since the variables $\lambda, \mu, \nu, \dots$ are continuous,⁵ there would have to be a time at which equality first occurs between one or more of the numerical values of $\lambda, \mu, \nu, \dots$ and the corresponding bound $l, m, n, \dots$, without any of the remaining variables having yet exceeded its bound. For this instant the negative value of $\varphi(\lambda, \mu, \nu, \dots)$

would not be numerically smaller than p , and therefore the right-hand side of our equation, in view of the preceding inequality relating to the initial state, would be negative; this contradicts the nature of the left-hand side. At the same time it is clear that the velocities v will also always remain enclosed within bounds, since one always has

$$\Sigma mv^2 \leq \Sigma mv_0^2 - \varphi(\lambda_0, \mu_0, \nu_0, \dots),$$

³Traité de Mécanique, tome second, page 492.

⁴This citation is absent in the Academy report. K.

⁵The words "and since the variables $\lambda, \mu, \nu, \dots$ are continuous" are absent in the Academy report. K.

and also that the bounds for each v , as well as for each of the variables $\lambda, \mu, \nu, \dots$ determining the position of the system, can be made arbitrarily narrow, since $l, m, n, \dots$ are capable of every degree of smallness.⁶

Finally, we wish to point out a remarkable error, found in various writers, concerning the subject just discussed.⁷ It is asserted that if the system passes through several equilibrium positions in the course of its motion, then these are alternately positions of stable and unstable equilibrium, that is, positions to which a maximum or a minimum of the function $\varphi(\lambda, \mu, \nu, \dots)$ corresponds. This assertion is based, first, on the fact that a maximum or minimum does not lose this character when the variables $\lambda, \mu, \nu, \dots$ formerly regarded as independent obtain the special values for which such a maximum or minimum occurs by becoming functions of a new quantity t ; and second, on the fact that for a function of a single variable t , maxima and minima follow one another alternately. Both statements are correct, but they do not justify the conclusion drawn from them. Rather, it would be necessary for the first statement also to remain valid conversely: that is, that the function $\varphi(\lambda, \mu, \nu, \dots)$ depending only on t , if it presents a maximum or minimum for a particular t , should retain the same property when the corresponding values of $\lambda, \mu, \nu, \dots$ are regarded as special values of the quantities $\lambda, \mu, \nu, \dots$ considered as independent variables. This need not be the case, even if only one quantity, for example λ , occurs. Thus the function λ^2 has only a minimum and no maximum at all, whereas the function $\sin^2 t$, into which the former passes under the assumption $\lambda = \sin t$, has not only infinitely many minima, which are equal to one another and to the former one, but also infinitely many maxima. The error into which one has fallen in this respect must be all the more surprising since the incorrectness of the assertion made is already recognizable in the ordinary motion of the pendulum.

⁶Here the extract in the Academy report closes; cf. p. 37 in the 1846 volume. K.

⁷Traité de Mécanique par Poisson, tome second, page 491.